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# Problem Set 11

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## 1 Angular Momentum and Spherical Harmonics

1.

$$\begin{aligned}\hat{\nabla}^2 Y_{1,-1} &= \left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \left( \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta \right) \\ &= \frac{1}{2} \sqrt{\frac{3}{2\pi}} \left( \frac{e^{-i\varphi}}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \sin \theta \right) + \frac{1}{\sin \theta} \frac{\partial^2}{\partial \varphi^2} e^{-i\varphi} \right) \\ &= \frac{1}{2} \sqrt{\frac{3}{2\pi}} \left( \frac{e^{-i\varphi}}{\sin \theta} \frac{\partial}{\partial \theta} (\sin \theta \cos \theta) - \frac{e^{-i\varphi}}{\sin \theta} \right) \\ &= \frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{e^{-i\varphi}}{\sin \theta} (\cos^2 \theta - \sin^2 \theta - 1) \\ &= \frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{e^{-i\varphi}}{\sin \theta} (1 - 2 \sin^2 \theta - 1) \\ &= \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta \\ &= 2Y_{1,-1}.\end{aligned}$$

$$\begin{aligned}\hat{\nabla}^2 Y_{1,0} &= \left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \left( \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta \right) \\ &= \frac{1}{2} \sqrt{\frac{3}{\pi}} \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (-\sin^2 \theta) \\ &= -\frac{1}{2} \sqrt{\frac{3}{\pi}} \frac{1}{\sin \theta} 2 \sin \theta \cos \theta \\ &= -\sqrt{\frac{3}{\pi}} \cos \theta \\ &= \sqrt{\frac{3}{\pi}} \cos \theta \\ &= 2Y_{1,0}.\end{aligned}$$

$$\begin{aligned}
\hat{\nabla}^2 Y_{1,1} &= \left( \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2} \right) \left( -\frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin \theta \right) \\
&= -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \left( \frac{e^{i\varphi}}{\sin \theta} \frac{\partial}{\partial \theta} \left( \sin \theta \frac{\partial}{\partial \theta} \sin \theta \right) + \frac{1}{\sin \theta} \frac{\partial^2}{\partial \varphi^2} e^{i\varphi} \right) \\
&= -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{e^{i\varphi}}{\sin \theta} \left( \frac{\partial}{\partial \theta} \sin \theta \cos \theta - 1 \right) \\
&= -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{e^{i\varphi}}{\sin \theta} (-\sin^2 \theta + \cos^2 \theta - 1) \\
&= -\frac{1}{2} \sqrt{\frac{3}{2\pi}} \frac{e^{i\varphi}}{\sin \theta} (1 - 2\sin^2 \theta - 1) \\
&= \sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin \theta \\
&= 2Y_{1,1}.
\end{aligned}$$

It follows that

$$\frac{\hat{\nabla}^2 Y_{\ell,m}}{Y_{\ell,m}} = 2$$

for all given  $Y_{\ell,m}$ .

2.

$$\begin{aligned}
L_+ Y_{1,-1} &= \left( \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right) \right) \left( \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta \right) \\
&= \frac{\hbar e^{i\varphi}}{2} \sqrt{\frac{3}{2\pi}} (e^{-i\varphi} \cos \theta + e^{-i\varphi} \cot \theta \sin \theta) \\
&= \frac{\hbar}{2} \sqrt{\frac{3}{2\pi}} (\cos \theta + \cos \theta) \\
&= \hbar \sqrt{2} \sqrt{\frac{3}{\pi}} \cos \theta \\
&= \sqrt{2} \hbar Y_{1,0}.
\end{aligned}$$

$$\begin{aligned}
L_+ Y_{1,0} &= \left( \hbar e^{i\varphi} \left( \frac{\partial}{\partial \theta} + i \cot \theta \frac{\partial}{\partial \varphi} \right) \right) \left( \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta \right) \\
&= \frac{\hbar e^{i\varphi}}{2} \sqrt{\frac{3}{\pi}} (-\sin \theta) \\
&= -\frac{\hbar}{2} \sqrt{\frac{3}{\pi}} e^{i\varphi} \sin \theta \\
&= -\hbar \sqrt{2} \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin \theta \\
&= \hbar \sqrt{2} Y_{1,1}.
\end{aligned}$$

$$\begin{aligned}
L_- Y_{1,0} &= \left( \hbar e^{-i\varphi} \left( -\frac{\partial}{\partial \theta} + i \cot \frac{\partial}{\partial \varphi} \right) \right) \left( \frac{1}{2} \sqrt{\frac{3}{\pi}} \cos \theta \right) \\
&= \frac{\hbar e^{-i\varphi}}{2} \sqrt{\frac{3}{\pi}} (\sin \theta) \\
&= \frac{\hbar}{2} \sqrt{\frac{3}{\pi}} e^{-i\varphi} \sin \theta \\
&= \hbar \sqrt{2} \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta \\
&= \hbar \sqrt{2} Y_{1,-1}.
\end{aligned}$$

$$\begin{aligned}
L_- Y_{1,1} &= \left( \hbar e^{-i\varphi} \left( -\frac{\partial}{\partial \theta} + i \cot \frac{\partial}{\partial \varphi} \right) \right) \left( -\frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{i\varphi} \sin \theta \right) \\
&= -\frac{\hbar e^{-i\varphi}}{2} \sqrt{\frac{3}{2\pi}} \left( -e^{i\varphi} \cos \theta - \frac{\cos \theta}{\sin \theta} e^{i\varphi} \sin \theta \right) \\
&= -\frac{\hbar}{2} \sqrt{\frac{3}{2\pi}} (-\cos \theta - \cos \theta) \\
&= \hbar \sqrt{\frac{3}{2\pi}} \cos \theta \\
&= 2\hbar Y_{1,0}.
\end{aligned}$$

3. Recall that we took the  $Y_{\ell,ms}$  to be orthonormal. By the previous problem, we have

$$\begin{aligned}
L_+ &\cong \begin{pmatrix} \langle Y_{1,1} | L_+ | Y_{1,1} \rangle & \langle Y_{1,1} | L_+ | Y_{1,0} \rangle & \langle Y_{1,1} | L_+ | Y_{1,-1} \rangle \\ \langle Y_{1,0} | L_+ | Y_{1,1} \rangle & \langle Y_{1,0} | L_+ | Y_{1,0} \rangle & \langle Y_{1,0} | L_+ | Y_{1,-1} \rangle \\ \langle Y_{1,-1} | L_+ | Y_{1,1} \rangle & \langle Y_{1,-1} | L_+ | Y_{1,0} \rangle & \langle Y_{1,-1} | L_+ | Y_{1,-1} \rangle \end{pmatrix} \\
&= \begin{pmatrix} 0 & \langle Y_{1,1} | \hbar | Y_{1,1} \rangle & \langle Y_{1,1} | 2\hbar | Y_{1,0} \rangle \\ 0 & \langle Y_{1,0} | \hbar | Y_{1,1} \rangle & \langle Y_{1,0} | 2\hbar | Y_{1,0} \rangle \\ 0 & \langle Y_{1,-1} | \hbar | Y_{1,1} \rangle & \langle Y_{1,-1} | 2\hbar | Y_{1,0} \rangle \end{pmatrix} \\
&= \hbar \sqrt{2} \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}.
\end{aligned}$$

$$\begin{aligned}
L_- &\cong \begin{pmatrix} \langle Y_{1,1} | 2\hbar | Y_{1,0} \rangle & \langle Y_{1,1} | \hbar | Y_{1,-1} \rangle & 0 \\ \langle Y_{1,0} | 2\hbar | Y_{1,0} \rangle & \langle Y_{1,0} | \hbar | Y_{1,-1} \rangle & 0 \\ \langle Y_{1,-1} | 2\hbar | Y_{1,0} \rangle & \langle Y_{1,-1} | \hbar | Y_{1,-1} \rangle & 0 \end{pmatrix} \\
&= \hbar \sqrt{2} \begin{pmatrix} 0 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.
\end{aligned}$$

With this, we obtain

$$L_x = \frac{\hbar \sqrt{2}}{2} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

$$L_y = \frac{-i\hbar\sqrt{2}}{2} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix}.$$

Note that

$$L_z |Y_{1,-1}\rangle = -i\hbar \frac{\partial}{\partial \varphi} \frac{1}{2} \sqrt{\frac{3}{2\pi}} e^{-i\varphi} \sin \theta = -\hbar |Y_{1,-1}\rangle,$$

$$L_z |Y_{1,0}\rangle = -i\hbar \frac{\partial}{\partial \varphi} \frac{1}{2} \sqrt{\frac{3}{2\pi}} \cos \theta = 0.$$

We are given that  $L_z |Y_{1,1}\rangle = \hbar |Y_{1,1}\rangle$ . In fact we can just say  $L_z |Y_{\ell,m}\rangle = \hbar m |Y_{\ell,m}\rangle$  so I don't know why I did this. This gives us

$$L_z = \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}.$$

4. Since  $L_+$  raises  $m$ , and the only other  $m \leq 1$  it can raise is 0, we must have that  $\langle Y_{1,1} | L_+ | Y_{1,0} \rangle$  is nonzero. We have that  $L_+ |Y_{1,0}\rangle = d |Y_{1,1}\rangle$  for  $d$  some proportionality constant.

Note that

$$d = d \langle Y_{1,1} | Y_{1,1} \rangle = \langle Y_{1,1} | L_+ | Y_{1,0} \rangle \implies d^\dagger = (\langle Y_{1,1} | L_+ | Y_{1,0} \rangle)^\dagger = \langle Y_{1,0} | L_- | Y_{1,1} \rangle.$$

I don't know math that allows me to do what I'm about to do, because  $L_-$  is not Hermitian, so technically it is supposed to only act on the  $\langle Y_{1,0} |$  in the above expression according to the definition of an adjoint. I don't have time to figure out the proper way to do this. We thus have

$$d = \langle Y_{1,0} | L_- | Y_{1,1} \rangle = \langle Y_{1,0} | k | Y_{1,0} \rangle = k$$

for  $k$  the proportionality constant  $L_- |Y_{1,1}\rangle = k |Y_{1,0}\rangle$ . Therefore, we can compute as follows.

$$|L_- |Y_{1,1}\rangle|^2 = |d|^2 \langle Y_{1,0} | Y_{1,0} \rangle = |d|^2 \langle Y_{1,1} | L_+ L_- | Y_{1,1} \rangle = |d|^2.$$

Note that

$$\begin{aligned} \langle Y_{1,0} | L_+ L_- | Y_{1,0} \rangle &= \langle Y_{1,1} | L_+ L_- | Y_{1,1} \rangle \\ &= \langle Y_{1,1} | L_+ L_- | Y_{1,1} \rangle + 0 \\ &= \langle Y_{1,1} | L_+ L_- | Y_{1,1} \rangle + \langle Y_{1,0} | L_- L_+ | Y_{1,1} \rangle \\ &= \langle Y_{1,1} | [L_+, L_-] | Y_{1,1} \rangle \\ &= \langle Y_{1,1} | 2\hbar L_z | Y_{1,1} \rangle \\ &= 2\hbar^2 \langle Y_{1,1} | Y_{1,1} \rangle \\ &= 2\hbar^2. \end{aligned}$$

Choosing  $d$  to be real and positive, we have  $d = \sqrt{2}\hbar$ . We then have

$$L_+ = \begin{pmatrix} 0 & \hbar\sqrt{2} & 0 \\ 0 & 0 & \hbar\sqrt{2} \\ 0 & 0 & 0 \end{pmatrix} \implies L_-^\dagger = \begin{pmatrix} 0 & 0 & 0 \\ \hbar\sqrt{2} & 0 & 0 \\ 0 & \hbar\sqrt{2} & 0 \end{pmatrix}.$$

5. Define

$$\left| \frac{1}{2}, \frac{1}{2} \right\rangle \cong \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \cong \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

Note that  $L_+$  will thus take the form

$$\begin{pmatrix} \left\langle \frac{1}{2}, \frac{1}{2} \right| L_+ \left| \frac{1}{2}, \frac{1}{2} \right\rangle & \left\langle \frac{1}{2}, \frac{1}{2} \right| L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \\ \left\langle \frac{1}{2}, -\frac{1}{2} \right| L_+ \left| \frac{1}{2}, \frac{1}{2} \right\rangle & \left\langle \frac{1}{2}, -\frac{1}{2} \right| L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \end{pmatrix}.$$

Since  $\left| \frac{1}{2}, \frac{1}{2} \right\rangle$  cannot be raised, clearly the entire left column will go to 0. We also have

$$\left\langle \frac{1}{2}, \pm \frac{1}{2} \right| L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \propto \left\langle \frac{1}{2}, \pm \frac{1}{2} \right| \frac{1}{2}, \frac{1}{2} \right\rangle = \begin{cases} 1, & \pm = + \\ 0, & \pm = - \end{cases}.$$

Ignore the cursed notation. This means we only need to compute a single entry, the one corresponding to

$$\left\langle \frac{1}{2}, \frac{1}{2} \right| L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle.$$

Let  $c$  give the proportionality constant:

$$\left| L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle \right|^2 = |c|^2 \left\langle \frac{1}{2}, \frac{1}{2} \right| \frac{1}{2}, \frac{1}{2} \right\rangle = \left\langle \frac{1}{2}, -\frac{1}{2} \right| L_- L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle$$

by the same trick as earlier. By the same logic as earlier, we once again have

$$\left\langle \frac{1}{2}, -\frac{1}{2} \right| L_- L_+ \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \left\langle \frac{1}{2}, -\frac{1}{2} \right| [L_-, L_+] \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \left\langle \frac{1}{2}, -\frac{1}{2} \right| -2\hbar L_z \left| \frac{1}{2}, -\frac{1}{2} \right\rangle = \hbar^2.$$

Therefore,  $c = \hbar$ , and

$$L_+ = \begin{pmatrix} 0 & \hbar \\ 0 & 0 \end{pmatrix}.$$

It follows that

$$L_- = L_+^\dagger = \begin{pmatrix} 0 & 0 \\ \hbar & 0 \end{pmatrix}.$$

We thus have

$$L_x = \frac{1}{2} \begin{pmatrix} 0 & \hbar \\ \hbar & 0 \end{pmatrix} = S_x, \quad L_y = \frac{-i}{2} \begin{pmatrix} 0 & \hbar \\ -\hbar & 0 \end{pmatrix} = S_y.$$

## 2 Angular Momentum and Born's Rule

1. The eigenvalues of  $L^2$  are  $\hbar^2 \ell(\ell + 1)$  for each corresponding  $\ell$ , meaning we can measure any of  $\{2\hbar^2, 6\hbar^2\}$  when  $\ell$  is 1 and 2, respectively. The probability of each component state being measured is the norm squared of it's coefficient, and the probability of each measurement is the sum over the probabilities of all states that yield taht specific eigenvalue. This gives us

$$\mathcal{P}_{\ell=1} = \left| \frac{1}{\sqrt{10}} \right|^2 + \left| -\frac{2}{\sqrt{10}} \right|^2 = \frac{1}{10} + \frac{4}{10} = \frac{1}{2},$$

$$\mathcal{P}_{\ell=2} = \left| \frac{2i}{\sqrt{10}} \right|^2 + \left| \frac{i}{\sqrt{10}} \right|^2 = \frac{4}{10} + \frac{1}{10} = \frac{1}{2}.$$

2. Each  $|\ell, m\rangle$  is an eigenvector of  $L_z$  with eigenvalue  $m$ . We thus have that the possible values are precisely  $\hbar$  times the  $m$  values, meaning we can measure  $\{0, \hbar, 2\hbar\}$ . Using By the same logic as part 1, the probabilities are

$$\mathcal{P}_{m=0} = \left| -\frac{2}{\sqrt{10}} \right|^2 + \left| \frac{i}{\sqrt{10}} \right|^2 = \frac{1}{2},$$

$$\mathcal{P}_{m=1} = \left| \frac{1}{\sqrt{10}} \right|^2 = \frac{1}{10},$$

$$\mathcal{P}_{m=2} = \left| \frac{2i}{\sqrt{10}} \right|^2 = \frac{2}{5}.$$

3. Histogram:

